

RESEARCH JOURNAL - 2026

Pravin Sharma
5th Year, Ph.D. Student

Department of Applied Science
William & Mary

2026

Table of Contents

1	March	1
1.1	Week - 12	1
1.2	Week - 13	4
1.3	Week - 14	7
2	April	8
2.1	Week - 14 - continued	8
	References	9

1 | March

This section presents the details of the task related to my research, which I completed in March.

1.1 Week - 12

Mar 15 - Mar 21

Week - 12

1. **Work on the Research only until Thursday and then for the Notes for Force on Friday & Saturday.**
2. Complete the first draft of the paper completely by the end of the week.
3. Work on your self-discipline and research only by spending at least 30 minutes reading Brian Tracy's book.
4. Make the file for PSSW calculation and submit the jobs for the same by Thursday.
5. Complete at least one note for the website [The Force one].
6. Submit the PSSW simulations and make the optimized script for the analysis.

Sunday Mar 15, 2026

To do Tasks:

1. Go to ISKCON and spend a good time with your wife there, and roam a little bit around Boston. ✓
2. Talk to Suwash and Shubh, and take proper rest. ✓

Focused Tasks of the day

→ Worked on the Prompt Engineering Course and tried to complete as much as possible.

Monday Mar 16, 2026

To do Tasks:

1. Complete the remaining part of the Prompt Engineering Course.
2. Work on the details for the writing of the hysteresis and $M(T)$ curve plots for the paper in $\LaTeX$.

Focused Tasks of the day

- Prompt Engineering Course for Gen-AI completion. ✓
- Work on hysteresis and $M(T)$ curve details for the paper after lunch.

Notes:

1. Work on the PhD research only throughout the day.

Tuesday Mar 17, 2026**To do Tasks:**

1. Go to the Burlington lab in the morning with Rajnandini and stay there till 4:30 pm. ✓
2. Attend the Lecture at Burlington. ✓
3. Gather the information on the details of the T_c , χ , and $M - H$ loop calculations from the VS Code and then put them in the Result section. ✗
4. Complete the writing there for T_c and χ . ✓
5. Think of the schematic and also the introductory part. ✗

Focused Tasks of the day

- Read the paper and files that you have used for the simulations and take notes on them.
- Attend the lecture and then go for lunch.

Notes:

- 1.

Wednesday Mar 18, 2026**To do Tasks:**

1. Complete the equilibrium properties writings.
2. Spend some time later on the introduction part.
3. Complete the Word document with the latest changes, including the references, and then email Prof. to have a look at it, even when the schematic is left.
4. Email the Library personnel and arrange the meeting next week related to the open access publishing.

Focused Tasks of the day

- Complete the hysteresis, T_c , and χ_m details and see where it can be added. ✓
- Attend the meeting with Emmanuel at 1:00 pm. ✓
- Complete the Word file with these changes. ✗
- Think of the introduction part and make sure to have some points before bed.

Notes:

1. Was not able to complete the writing because of personal issues. I need to learn how to manage those so I don't jeopardize my career.

Thursday Mar 19, 2026 - Rest and Reflection Day**To do Tasks:**

1. **NO ANY WORK TODAY, TAKE REST, ENJOY, GIVE YOURSELF A REFLECTION TIME**

Notes:

1. **REST DAY, REFLECTION DAY**

Friday Mar 20, 2026**To do Tasks:**

1. Complete the Word file with these changes.
2. Make the Excel sheet for the Job Applications and other details related to that.

Focused Tasks of the day

- Complete the writing first. **X**
- Complete the Word document. **✓**
- Email to the Librarian for the open publication related. **X**

Notes:

1. Did some writing and then transferred the details to a Word document. Need to complete this by Monday morning.

Saturday Mar 21, 2026**To do Tasks:**

1. Spend Time on self.
2. Complete 2 hours of LinkedIn Python training.
3. Start packing the bags for departure.
4. Go to ISKCON later in the evening to meet Rajnandini there and roam around a little bit.

Focused Tasks of the day

→

Notes:

1.

1.2 Week - 13

Mar 22 - Mar 28

Week - 13

1. Get the PSSW simulation plots completed after the Wednesday meeting.
2. Complete the schematic and introduction part of the paper [Thu-whole day]
3. Create an Excel Sheet for the job and its requirements, and get enrolled in the iTech Program.[Fri]
4. Email Uzaiiah, and that Jackson High School professor related to the High School Physics Jobs, and also other representatives whom I have met at the Job Fairs. [Fri-before lunch].
5. Work on completing the teaching philosophy writing and cover page writing.[Fri]

Sunday Mar 22, 2026**Notes:**

1. Spent most of the time talking and staying with wife.

Monday Mar 23, 2026**Notes:**

1. Spent the time packing and enjoying time with my wife.
2. Left from Boston with Shubh in the car and then reached his home in NJ, and he dropped me off at the Philadelphia Amtrak station for my train.

Tuesday Mar 24, 2026**To do Tasks:**

1. Work on the paper introduction. Read the VAMPIRE papers related to $\text{Co}_{50}\text{Fe}_{50}$. ✗

Notes:

1. Wasn't able to do any work because I was tired from the travel, so I spent the time resting and then preparing for the week.

Wednesday Mar 25, 2026

To do Tasks:

1. Go to the department after getting the Word file completed. ✓
2. Work on the schematic for the paper. ✗
3. Email the Library personnel for the open access thing, and also how this process goes along. Schedule a meeting and see how this is done. ✓
4. Arranged the stuff in the Calendar. ✓
5. Do the Updates to the website. ✓

Focused Tasks of the day

- Complete the Word file before you go to the department, leaving the introduction part behind.
- Work on the format to write the Introduction file.

Notes:

1. Go for a walk.

Thursday Mar 26, 2026

To do Tasks:

1. Complete some reading of the Atomic Habits Book. ✓
2. Go for a run. ✓
3. Read the different papers of VAMPIRE and other papers to get some more ideas and start some basic points on the introduction section of the paper. ✓
4. Go to Aago and spend time there; leave early, as I also need to go to Costco. ✓

Focused Tasks of the day

- Complete the main points that need to be included in the Introduction part of the Paper.

Notes:

1. I skipped the BCM potluck today, and I think I have made the correct decision based on this.

Friday Mar 27, 2026

To do Tasks:

1. Work on the writing maximum. ✗

2. Create an Excel sheet to update the job applications and collect maximum information for the jobs.
3. Make three different CVs that are suitable for;
 - High School Teaching.
 - Community College Teaching.
 - Adjunct Physics Faculty Teaching.
4. Update the iTeach things and email them.
5. Create the PSSW files and possibly run some demo calculations. ✗

Focused Tasks of the day

- Log in to the iTeach thing and then email the questions, if any, related to the provisional certificate things.
- Search for jobs and then fill the Excel sheet where you can apply for the jobs.
- Looking for jobs and putting the details of applications and the status of all.

Notes:

1. Went to the MSA event in Sadler Center.

Saturday Mar 28, 2026

To do Tasks:

1. Go for a long run in the morning. ✓
2. Attend the ANPA PER Monthly talk. ✗
3. Do the Tax Filing online with Rajnandini at 12:10 pm. [Not able to complete her taxes as the state income was more than the federal, and this was due to the issue of the address.] ✓
4. Complete the CV for teaching. ✗
5. Complete the Force Note Writing. ✗
6. Complete that 1.5 hrs Python training and also try to play guitar if Sadie is not in the house later. ✗

1.3 Week - 14

Mar 29 - Mar 31

Week - 14

1. Complete the unfinished task that was not completed yesterday.
2. Spend most of the week on JOB-related things.
3. Try to complete the Introduction part of the paper and share that to Professor by Thursday.
4. Be more action-oriented this week and submit the first draft of the paper.
- 5.

Sunday Mar 29, 2026

To do Tasks:

1. Go for Bee & Goody 5K run.
2. Complete the CV for teaching.
3. Work on the LinkedIn Learning course for Python before going to the Bee & Goody run.
4. Email the iTeach Personnel about the Certification verification and next processes.
5. Email the Walter John HS principal about the Physics teaching Jobs.

Monday Mar 30, 2026

To do Tasks:

1. Work on writing the introduction from the morning.
- 2.

Tuesday Mar 31, 2026

To do Tasks:

- 1.

* * *

2 | April

This section presents the details of the task related to my research, which I completed in April.

2.1 Week - 14 - continued

Apr 01 - Apr 04

Week - 14

1.

Wednesday Apr 01, 2026

To do Tasks:

1.

Thursday Apr 02, 2026

To do Tasks:

1.

Friday Apr 03, 2026

To do Tasks:

1.

Saturday Apr 04, 2026

To do Tasks:

1.

References